Computational Optics for Liquid Microsampling: a Multi-Channel Smartphone Sensing Platform

Svetoch Series, Paper V of V

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Abstract

We describe **MicroLab**, an application of the Svetoch method (Paper I) that turns an unmodified consumer smartphone into a multi-channel optical sensing platform for liquid microsamples. The OLED screen is used both as a controllable structured light source and — through its built-in matrix polarizer — as an opportunistic polarized source; the camera is the detector and the cover glass an optical element. A microdroplet on a disposable optical barrier is interrogated through three physically independent channels: **refractive** (a dispersion index $D = C_r/C_b$, focal spread and edge contrast, with temperature-drift correction; specific gravity, TDS, protein trends), **polarization** (Malus-law optical rotation of chiral analytes), and **absorption** (RGB spectrophotometry against an empty-lens baseline). A label-free **frustrated total internal reflection** mode uses the cover glass as an evanescent-wave biosensor. The channels are fused into one repeatable *optical fingerprint*. The core contribution is a **validation discipline**: a transfer-matrix self-QA gate (SVD rank/condition number) and a **Confidence Engine** that *rejects* a scan when signal is insufficient or droplet volume too small, instead of guessing. MicroLab is **research-use / wellness-and-food-tech, not a validated medical diagnostic**; medical claims (especially blood glucose) are explicitly **not** made.

Keywords: computational optics, smartphone diagnostics, refractometry, mobile polarimetry, spectrophotometry, frustrated total internal reflection, sensor fusion, confidence estimation, research-use-only.

1 Introduction

A modern smartphone packs, within millimetres, an emissive OLED display, an integrating camera, and a precision cover glass. Papers I–IV showed this hardware computes in light and senses its own physics. Paper V asks a practical question: can the same hardware extract a repeatable optical fingerprint from a single microdroplet of liquid?

The honest framing matters. We do *not* claim to "measure everything" in a drop; we claim the narrower, defensible goal of extracting **stable, physically grounded optical features** that are robust to droplet geometry and to hardware variability across phone models. The platform is architected around three independent optical channels whose fusion yields a fingerprint, plus a trust layer that knows when *not* to report a number. This validation discipline — not any individual algorithm — is the asset. We separate the platform into trust tiers: food and technical liquids (zero regulatory risk), wellness trends, and regulated medical claims (reserved for after clinical validation and cohort studies). Figure 1 shows the full signal stack.

Figure 14. MicroLab signal stack: physics-informed multi-channel fusion

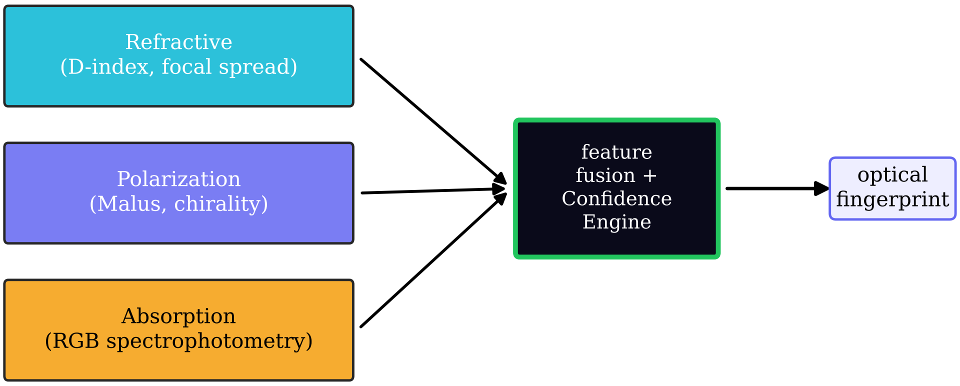

Figure 1: The MicroLab signal stack. Three physically independent channels — refractive (dispersion index, focal spread, edge contrast), polarization (Malus-law optical rotation), and absorption (RGB spectrophotometry) — feed a physics-informed feature-fusion layer guarded by the Confidence Engine, producing one repeatable optical fingerprint of the microdroplet.

2 The three-channel signal stack

The technical core is sensor **fusion**, not a single trick; each channel reports information the others cannot.

2.1 Refractive mode — dispersion index and interferometric path

The screen–air-gap–cover-glass–camera path is a refractometer and, with reference fringes, a Fabry–Pérot interferometer. The primary engineered feature is a **dispersion index**

$$D = \frac{C_r}{C_b}, \tag{1}$$

the ratio of red- to blue-channel response of light traversing the droplet. Because refractive index depends on wavelength and dissolved content, D tracks specific gravity, total dissolved solids and protein trends. Artefacts are averaged across multi-exposure and speckle patterns, and a mandatory temperature-drift layer $D_{\mathrm{corr}} = D - \kappa\,(T - T_0)$ removes thermo-optic drift from local screen heating. For transparent media the Fabry–Pérot fringe condition

$$2\,n\,d\,\cos\theta = m\,\lambda \tag{2}$$

gives, for an index change Δn, a fringe-phase shift $\Delta\phi = (4\pi d\cos\theta/\lambda)\,\Delta n$, measured to sub-pixel precision against the displayed reference pattern — optical density / concentration with no moving parts.

2.2 Polarization mode — Malus-law optical rotation

OLED panels carry a built-in polarizer to suppress ambient reflection, so emitted light is partially polarized; MicroLab treats it as a **free, opportunistic polarized source**. Analysing the angle-dependent intensity follows Malus's law,

$$I(\theta) = I_0 \cos^2(\theta - \alpha), \tag{3}$$

where α is the optical rotation of a chiral analyte. Fitting α versus path length and concentration yields a saccharimetry-style readout (sugars, glucose as R&D only), positioned as a research module providing a differentiation boost.

2.3 Absorption mode — RGB spectrophotometry

The OLED is a controllable RGB backlight and the camera a three-band photometer, applying the Beer–Lambert law,

$$A_\lambda = -\log_{10}\frac{I_\lambda}{I_{0,\lambda}} = \varepsilon_\lambda\, c\, \ell, \tag{4}$$

with $I_{0,\lambda}$ the intensity through an **empty-lens baseline**. Green-channel absorption tracks blood-like samples (haemoglobin); blue-channel attenuation tracks urine chromophore trends (urobilin). Baseline normalization makes the channel robust to backlight ageing and exposure changes.

3 Label-free frustrated-TIR sensing

A fourth, label-free modality uses the cover/prism glass as an **evanescent-wave biosensor**. Display light is coupled above the critical angle $\theta_c = \arcsin(n_2/n_1)$ for total internal reflection; the evanescent field extends a fraction of a wavelength into the sample. An analyte contacting the surface **frustrates** the reflection, and the reflected contrast/interference pattern read by the camera changes with the local refractive index and surface binding. Scanning the fringe period and recording the peak contrast ratio gives a quantitative response. Being interface- rather than bulk-sensitive, this mode complements the refractive and absorption channels and opens a route to surface-binding assays. The audit rates it (A3) as strong; it is disclosed here defensively as part of the open fingerprint stack.

4 Sensor self-QA by transfer-matrix SVD

No measurement is trustworthy without a metrology gate. Before any droplet, MicroLab characterizes the display→lens→sensor chain. Dark (null) frames and calibration patterns measure leakage and display→sensor cross-talk; one-hot display patterns build an empirical transfer matrix $\mathbf{M}$, whose singular-value decomposition

$$\mathbf{M} = \mathbf{U}\,\boldsymbol{\Sigma}\,\mathbf{V}^\top \tag{5}$$

yields the **rank** and **condition number** $\kappa = \sigma_{\max}/\sigma_{\min}$ — quantitative metrics of vignetting and optical degradation. A degraded or contaminated lens collapses the small singular values, raising κ and lowering the effective rank, and the channel is flagged unfit. This is the **"Auto-Air" quality gate**: an automatic baseline-aberration check run on air, before the sample, that must pass before measurement begins. Rated (B3) moderate; its value is operational — the first line of the validation discipline.

5 Feature fusion and the Confidence Engine

The channels do not vote in isolation. Physically grounded features — D and D_{corr}, focal spread, edge/structure contrast and standard deviation, the Malus rotation angle, per-band absorbances, and the frustrated-TIR contrast peak — are **fused** into one feature vector. The strategy is conservative: physics-informed engineered features first, a supervised model only on top, never raw pixels into a black box. The decisive component is the **Confidence Engine**, a trust layer that scores every scan and **rejects** it when evidence is inadequate (low SNR, too small a droplet, failed Auto-Air, cross-channel inconsistency). The product is expected to say *"not enough data, please re-scan"* rather than guess, and every result carries an explicit status (*estimated*, *trend-only*, *needs lab confirmation*). This honesty discipline is the moat: the contribution is the rejection logic and physically grounded fusion (A4, strong), not the classifier.

6 Measurement map and use-cases

The platform is built bottom-up, from low-regulation, easy-ground-truth scenarios toward medical trust (Table 1).

Table 1: Measurement map: targets, status and claim level.

Target	Status	Why promising	Claim level
Urine specific gravity	Strong MVP	Direct refractive link; minimal prep	Wellness → clinical (after vali
Coffee TDS / Brix (food)	Strong MVP	Zero regulatory risk; fast loop	Food-tech, immediate
Total serum protein	Research candidate	Strong optical tradition	Research-use-only
Saliva fertility (ovulation)	Conditional	Large consumer interest	Trend-assistance only
Tear osmolarity trend	Conditional	Niche premium market	Trend-only, later-stage
Blood glucose	High-risk R&D	Large market	**No claim** — R&D only

The strongest first products carry no medical risk: coffee TDS/Brix and water checks deliver revenue and diverse cross-device data immediately. Urine specific gravity is the natural bridge to a future clinical pilot (direct refractive link; cheap gold-standard refractometer). Blood glucose is retained only as an open research hypothesis with **no diagnostic claim**.

7 Methods

Hardware: reference device Xiaomi 12 Lite (6.55" AMOLED, 2400 × 1080, 120 Hz; 32 MP camera; built-in display polarizer; cover glass as TIR/refractive element). A disposable **optical barrier strip** sets sterility and a repeatable optical stack over the lens. **Ritual:** (1) apply barrier; (2) run the Auto-Air SVD gate before the droplet; (3) apply the droplet with an on-screen volume guide; (4) burst-capture while the screen emits the channel pattern sequences (refractive gradients, polarization sweep, RGB photometry, TIR fringe scan); (5) fuse features, run the Confidence Engine, report with status; (6) remove and discard the barrier. **Calibration / ground truth:** each mode has an external reference (refractometer, TDS meter, lab assay); datasets log phone model, screen type, ambient light, temperature, film thickness, and timestamped repeats for drift. **Software:** built on the Svetoch on-device stack (Paper I).

8 Discussion: hard truths

Confounders as measurable nuisance variables. Sample inhomogeneity, ambient-light leakage, barrier-film microcracks and local screen overheating are real; MicroLab treats them as measurable nuisance variables handled by quality gates (multi-exposure/speckle averaging, empty-lens baseline, Auto-Air, temperature-drift correction). Unbounded confounders trigger a rejection. **Regulatory tiers.** Claims are split into *educational*, *wellness*, *research-use-only*, and *regulated diagnostic*; the first three are available now, the last reserved strictly for after clinical validation and cohort studies (planned pilot: urine specific gravity in a clinic). **Explicit non-medical claims.** MicroLab as described is **not a validated medical diagnostic**; no diagnostic claim is made for any biofluid, and blood glucose is explicitly excluded from any present claim. The contribution is the *honesty and confidence discipline* — a system that refuses to guess. **Patent context:** the audit rates polarimetry (A2), frustrated-TIR (A3) and multimodal fusion with the Confidence Engine (A4) as strong, and the interferometer/refractometer (B2) and sensor self-QA (B3) as moderate; all are disclosed openly here.

9 Conclusion

A smartphone can be a multi-channel optical sensing platform. Combining a refractive dispersion channel, an opportunistic Malus-law polarization channel, RGB absorption spectrophotometry, and label-free frustrated-TIR sensing, MicroLab extracts a repeatable optical fingerprint from a single microdroplet. Guarded by a transfer-matrix self-QA gate and a Confidence Engine that rejects untrustworthy scans, it is honest about what it can and cannot measure. The validation discipline — not the machine learning — is the contribution, and the medical tier remains explicitly off-limits until clinical validation.

Data and code. github.com/infosave2007/svetoch; related VMF/NVG theory: github.com/infosave2007/vm (Apache-2.0).

Priority note. Released as a defensive publication to establish authorship and date of disclosure; the author has elected not to seek patent protection.

References

[1] B. Edlén, "The refractive index of air," *Metrologia* **2**, 71 (1966).

[2] E. Hecht, *Optics*, 5th ed., Pearson, 2017 (Malus's law; polarimetry/saccharimetry).

[3] D. F. Swinehart, "The Beer–Lambert law," *J. Chem. Educ.* **39**, 333 (1962).

[4] X. Fan et al., "Sensitive optical biosensors for unlabeled targets: a review," *Anal. Chim. Acta* **620**, 8 (2008).

[5] A. Ozcan and E. McLeod, "Lensless imaging and sensing," *Annu. Rev. Biomed. Eng.* **18**, 77 (2016).

[6] G. H. Golub and C. F. Van Loan, *Matrix Computations*, 4th ed., Johns Hopkins, 2013 (SVD / system characterization).

Part of the *Svetoch* series (defensive publication, not patented). Project and code: github.com/infosave2007/svetoch; related VMF/NVG theory: github.com/infosave2007/vmf. Released for the public record to establish authorship and priority.